LAB 2

GEOSPATIAL REPRESENTATION LEARNING

Lab 2: Geodata Scavenger Hunt

PRESS → OR SPACE. →

© Marc Rußwurm
Licensed under **Creative Commons Attribution–NonCommercial 4.0 International (CC BY-NC 4.0)**.
You may share and adapt these slides for teaching, research, and other non-commercial educational purposes with attribution.
Commercial training, paid workshops, consulting seminars, or incorporation into commercial course products require prior permission.

UNIVERSITÄT BONN

The Challenge

Find one way of measuring our area of interest on Earth.

Each of you will acquire one different geospatial data modality
and contribute it to our shared class dataset.

Same Place — Many Measurements

AOI →

- 📡 radar backscatter
- 🛰️ optical reflectance
- 🏔️ elevation
- 👥 population
- 🌊 soil moisture
- 🌡️ surface temperature
- 🌧️ precipitation
- 🌱 land cover
- 🛣️ road network
- *… and more*

The same coordinates can carry many different measurements. Each student contributes one layer.

Your Assignment

Every student receives:

- One assigned data modality or source
- The same Area of Interest (AOI)
- The same required output CRS

Work on your assigned source only.

Assignment table

⚠ *Instructor: insert or link to the student–dataset assignment table here.*

AOI: *[to be provided]*
Required output CRS: *[e.g. EPSG:XXXXX]*
Shared Drive folder: *[link]*
Shared Google Slides: *[link]*

What You Need to Deliver

① One geospatial data file covering the AOI

② Data kept at its **native spatial resolution** — do not resample

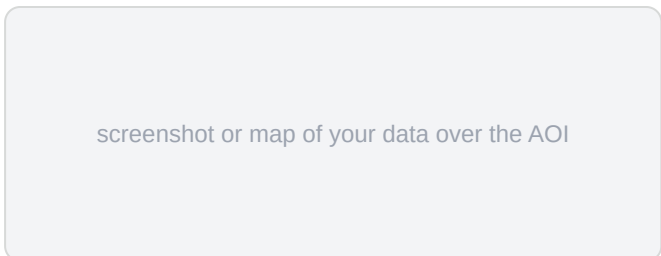
③ Data reprojected to the required output CRS

④ File uploaded to the shared Google Drive folder

⑤ One completed slide in the shared Google Slides deck

Your Slide (Template)

[Dataset / product name]



What does it measure? …
Provider / source …
Native spatial resolution …
Temporal reference …
Native CRS …
Final CRS …
File format …

How did you get it? (2–3 short steps)
One difficulty or surprise (one sentence)

The Workflow

Find → **Understand** → **Download** → **Clip** → **Reproject** → **Check** → **Upload**

Before downloading anything, determine:

- What does this dataset actually represent?
- What is its CRS?
- What is its spatial resolution?
- What is its temporal meaning?
- What are the nodata / missing-value conventions?

Coordinate Systems Matter

source dataset

native CRS

(e.g. geographic WGS 84)

↓

coordinate transformation

required output

class CRS

[EPSG:XXXXX]

Reprojection changes the **coordinate representation**.
It should not change the **spatial resolution**.

After reprojection: verify that the data still overlaps the AOI.

Plot it. Do not assume.

Native Resolution Is Intentional

Do not make the datasets look the same yet.

elevation ~30 m	**optical imagery** 10 m	**population** 100 m
land cover native categorical grid	**climate** kilometres	*your modality*

The resolution mismatch is part of the problem.

A Teaser for Lab 3

Continuous values

- elevation
- surface temperature
- spectral reflectance

interpolation between pixels is meaningful

Categorical values

- land-cover class
- soil type

interpolation between pixels is not meaningful

What should the value be between two pixels?
This depends on what the values mean.

If You Get Stuck

1. Read the dataset / product documentation
2. Check the CRS and spatial extent of the file you downloaded
3. Inspect the downloaded file locally — open it, plot it
4. Ask a neighbour what they would check first
5. Ask the instructor

Do not spend 45 minutes fighting a broken download portal.
If a source is inaccessible or genuinely broken, flag it early.

Timeline

00:00–00:10	Briefing and assignments
00:10–00:25	Find your source — understand the product and its metadata
00:25–01:10	Download / access the data
01:10–01:30	Clip to AOI, reproject to class CRS, verify
01:30–01:40	Upload data file and complete your slide
01:40–02:10	**Lightning presentations**

Lightning Presentations

60–90 seconds per student

Answer only:

1. What does your dataset measure?
2. Where did you get it?
3. What did you have to do to make it usable?
4. What was the hardest or most surprising part?

No deep dives into the science behind the modality.

No live demos unless everything already works.

The variety of acquisition stories — APIs, portals, registration walls, 4 GB files — is itself the learning content.

What We Built Together

one place

AOI

+

N independent modalities

your datasets

=

shared result

multimodal geospatial dataset

Keep your data and code. We will use them again in Lab 3.

Next step: coordinates → sampled values → machine-learning dataset